Enterprise Corporate Office Network

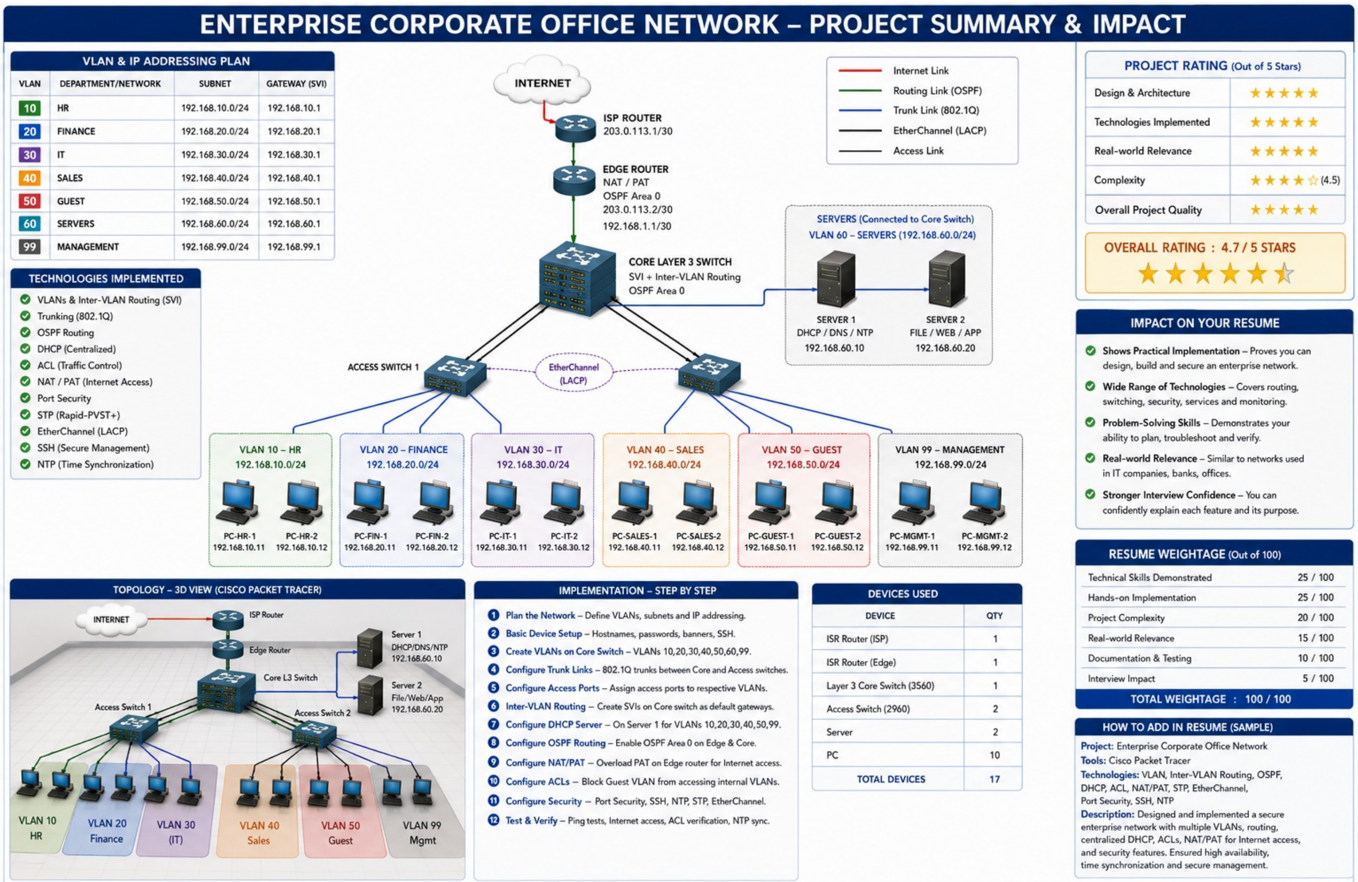
Complete Project Documentation – Cisco Packet Tracer

Purpose: This document is a handover/reference guide. A new person should be able to understand the topology, IP plan, device roles, traffic flow, services, security, testing and troubleshooting without needing the original presentation.

1. Project Overview

This project is a simulated enterprise corporate network designed in Cisco Packet Tracer. It separates departments using VLANs, provides inter-VLAN routing through a Layer-3 Core Switch, uses OSPF for dynamic routing, centralized DHCP/DNS/NTP services, NAT/PAT for Internet access, ACLs for traffic control, and security/redundancy features such as SSH, Port Security, STP and LACP EtherChannel.

2. Complete Topology



Internet → ISP Router → Edge Router → Core Layer-3 Switch → Access Switches → Department PCs
Core Layer-3 Switch → Server VLAN → Server 1 / Server 2

3. Device Roles

Device	Qty	Role	Key Address / Note
ISP Router	1	Represents ISP/Internet edge	203.0.113.1/30 shown
Edge Router	1	NAT/PAT, OSPF, Internet gateway	203.0.113.2/30; 192.168.1.1/30
Layer-3 Core Switch (3560)	1	SVIs, inter-VLAN routing, OSPF, central aggregation	VLAN gateways .1
Access Switch (2960)	2	User access, VLANs, trunks, EtherChannel, STP	Connects PCs
Server	2	DHCP/DNS/NTP and File/Web/App	192.168.60.10 / .20
PC	10	Department and management endpoints	VLAN 10/20/30/40/50/99

4. VLAN and IP Addressing Plan

All user/server VLANs use /24 networks. The .1 address is the SVI/default gateway for each VLAN.

VLAN	Department	Subnet	Gateway	Endpoints
10	HR	192.168.10.0/24	192.168.10.1	PC-HR-1 .11, PC-HR-2 .12
20	Finance	192.168.20.0/24	192.168.20.1	PC-FIN-1 .11, PC-FIN-2 .12
30	IT	192.168.30.0/24	192.168.30.1	PC-IT-1 .11, PC-IT-2 .12
40	Sales	192.168.40.0/24	192.168.40.1	PC-SALES-1 .11, PC-SALES-2 .12
50	Guest	192.168.50.0/24	192.168.50.1	PC-GUEST-1 .11, PC-GUEST-2 .12
60	Servers	192.168.60.0/24	192.168.60.1	Server 1 .10, Server 2 .20
99	Management	192.168.99.0/24	192.168.99.1	PC-MGMT-1 .11, PC-MGMT-2 .12

5. Endpoint IP Addressing

Device	VLAN	IP Address	Default Gateway
PC-HR-1	10	192.168.10.11	192.168.10.1
PC-HR-2	10	192.168.10.12	192.168.10.1
PC-FIN-1	20	192.168.20.11	192.168.20.1
PC-FIN-2	20	192.168.20.12	192.168.20.1
PC-IT-1	30	192.168.30.11	192.168.30.1
PC-IT-2	30	192.168.30.12	192.168.30.1
PC-SALES-1	40	192.168.40.11	192.168.40.1
PC-SALES-2	40	192.168.40.12	192.168.40.1
PC-GUEST-1	50	192.168.50.11	192.168.50.1
PC-GUEST-2	50	192.168.50.12	192.168.50.1
PC-MGMT-1	99	192.168.99.11	192.168.99.1
PC-MGMT-2	99	192.168.99.12	192.168.99.1

6. Server Plan

Server	IP	VLAN	Services	Purpose
Server 1	192.168.60.10	60	DHCP / DNS / NTP	Central network services
Server 2	192.168.60.20	60	File / Web / App	Application and file services

7. WAN / Routing Addressing

Link	Network	Device A	Device B	Purpose
ISP ↔ Edge	203.0.113.0/30	ISP: 203.0.113.1	Edge: 203.0.113.2	Internet-facing point-to-point link
Edge ↔ Core	192.168.1.0/30	Edge: 192.168.1.1	Core: 192.168.1.2	Internal routed link / OSPF

8. How the Network Works

A. PC to PC in different VLANs: Source PC checks the destination, sends the frame to its default gateway (SVI), and the Core Layer-3 Switch routes it to the destination VLAN.

B. PC to Internet: PC → Access Switch → Core SVI → Edge Router → NAT/PAT → ISP Router → Internet. Return traffic is translated back by the Edge Router.

C. DHCP: A client broadcasts a DHCP request. The VLAN SVI uses DHCP relay (ip helper-address) to forward the request to Server 1 at 192.168.60.10. The server returns the address information to the client.

D. Routing: The Core and Edge exchange routes using OSPF Area 0. This removes the need to maintain every internal route manually.

E. Guest security: Guest VLAN 50 is separated from corporate VLANs. ACL policy is used to restrict unwanted access to internal networks while allowing the intended services/Internet access.

9. Switching Design

Access ports: End-user PCs are placed in their respective VLANs.

802.1Q trunks: Core-to-Access uplinks carry the required VLANs between switches.

EtherChannel with LACP: Multiple physical uplinks are bundled into a logical Port-Channel, increasing resilience and aggregate capacity.

STP/PVST+: Provides Layer-2 loop prevention where redundant switching paths exist.

10. Layer-3 Design

SVIs: VLAN 10/20/30/40/50/60/99 use .1 as their gateway on the Core Layer-3 Switch.

OSPF Area 0: The Edge Router and Core exchange routes across the 192.168.1.0/30 routed link.

Default/Internet path: The Edge Router is the boundary toward the ISP and performs NAT/PAT for inside private networks.

11. Services Design

Service	Location	What it does
DHCP	Server 1 – 192.168.60.10	Assigns IP address, mask, gateway and DNS information to clients.
DNS	Server 1 – 192.168.60.10	Resolves host/domain names to IP addresses.
NTP	Server 1 – 192.168.60.10	Synchronizes device clocks for reliable logs and troubleshooting.
File/Web/App	Server 2 – 192.168.60.20	Provides application/file/web services for the enterprise.

12. Security Design

- **VLAN segmentation:** separates departments and the Guest network.
- **ACL:** controls which networks/services can communicate.
- **Port Security:** helps prevent unauthorized devices on access ports.
- **SSH:** secure encrypted management instead of Telnet.
- **Management VLAN 99:** provides a dedicated network for management endpoints.

13. Implementation Sequence

1. Plan VLANs, subnets, gateways and device roles.

2. Configure hostnames, basic device settings and management/SSH.

3. Create VLANs on the switching infrastructure.

4. Configure access ports for the correct department VLANs.

5. Configure Core–Access 802.1Q trunks.

6. Configure EtherChannel/LACP on the redundant uplinks.

7. Configure STP/PVST+ and verify the Layer-2 topology.

8. Configure Core SVIs and enable Layer-3 routing.

9. Configure DHCP server scopes and DHCP relay on the relevant SVIs.

10. Configure OSPF Area 0 between Core and Edge.

11. Configure NAT/PAT on the Edge Router for Internet access.

12. Apply ACL policies, especially for Guest traffic.

13. Configure DNS/NTP and server services.

14. Configure Port Security on appropriate access ports.

15. Test end-to-end connectivity and troubleshoot failures.

14. Verification Checklist

• show vlan brief – verify VLAN creation and access-port membership.

• show interfaces trunk – verify 802.1Q trunking and allowed VLANs.

• show etherchannel summary – verify LACP/Port-Channel status.

• show spanning-tree – verify STP state and VLAN root/blocked paths.

• show ip interface brief – verify SVI/interface status and IPs.

• show ip route – verify connected, OSPF and default routes.

• show ip ospf neighbor – verify OSPF adjacency.

• show ip ospf interface brief – verify OSPF-enabled interfaces.

• show ip nat translations / show ip nat statistics – verify NAT/PAT.

• show access-lists – verify ACL matches and counters.

• show port-security interface – verify port-security status.

• show ip dhcp binding – verify DHCP leases where applicable.

• ping between same-VLAN and different-VLAN PCs.

• Ping the server IPs 192.168.60.10 and 192.168.60.20.

• Test DNS/NTP and Internet reachability from permitted clients.

15. Troubleshooting Flow

PC has no IP: Check access VLAN → SVI → DHCP relay → server reachability → DHCP scope.

Same VLAN works but different VLAN fails: Check SVI status, ip routing, gateway and Core routing table.

OSPF not working: Check both interfaces are up, IPs/masks match, OSPF network/interface configuration and neighbor state.

Internet fails but internal ping works: Check Edge default route → NAT inside/outside → NAT ACL/match → ISP reachability.

Trunk problem: Check trunk mode, allowed VLANs, native VLAN consistency and VLAN existence.

EtherChannel problem: Check that member interfaces have matching speed/duplex and identical channel configuration.

Guest cannot reach intended destination: Check ACL direction, ACL order and counters.

16. End-to-End Test Scenarios

Test	Expected Result
HR PC → HR PC	Same VLAN communication succeeds.
HR PC → Finance PC	Inter-VLAN routing succeeds if ACL permits it.
User PC → Server 1	DHCP/DNS/NTP server is reachable if policy permits.
User PC → Server 2	File/Web/App server is reachable if policy permits.
Guest → Corporate VLAN	Blocked where ACL policy denies internal access.
User PC → Internet	Works through Edge Router NAT/PAT if permitted.
Core ↔ Edge	OSPF neighbor establishes and routes are learned.
Access ↔ Core	Trunk and EtherChannel remain operational.

17. Interview / Handover Summary

“This is an enterprise corporate network built in Cisco Packet Tracer. I segmented departments using VLANs, used a Layer-3 Core Switch for SVI-based inter-VLAN routing, connected access switches using 802.1Q trunks and LACP EtherChannel, and used STP for loop prevention. OSPF Area 0 provides dynamic routing between the Core and Edge Router. A centralized server provides DHCP, DNS and NTP, while the Edge Router performs NAT/PAT for Internet access. ACLs, Port Security and SSH add traffic control and secure management. The network was tested with connectivity, routing, DHCP, NAT, ACL and service verification.”

18. Important Documentation Note

The supplied topology image gives the device types, VLAN/IP plan, services and link roles. Exact physical interface numbers (for example Gi0/1, Fa0/1, etc.), exact DHCP pool names, ACL sequence numbers, SSH usernames/passwords and some command-level settings are not visible in the image. Those should be taken directly from the original .pkt file if you need an exact command-by-command configuration record. This document therefore avoids inventing interface numbers or credentials.